

Data-driven control of linear systems using quantized data

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Abstract—This paper studies data-driven stabilization of unknown discrete-time linear systems using only quantized state measurements that take values in finite sets. The proposed approach consists of two stages. In the controller design stage, we collect quantized data while maintaining a prescribed quantization error bound and use them to formulate a semidefinite program (SDP). A verifiable condition is established under which any feasible solution to the SDP yields a stabilizing feedback gain. In the stabilization stage, a Lyapunov-based quantizer update rule is developed to guarantee exponential convergence under quantized state feedback. As a key feature, the number of quantization cells remains finite in both stages and constant during stabilization. Simulation results illustrate the effectiveness of the proposed approach.

I. INTRODUCTION

Data-driven methods provide powerful tools for controlling systems where accurate models are unavailable or difficult to obtain [1], [2]. Indirect approaches based on system identification followed by model-based control design may suffer from model mismatch, which can compromise closed-loop stability and performance guarantees. In contrast, direct data-driven methods bypass explicit model identification and design controllers directly from data [3], [4]. A key development in this direction is Willems' fundamental lemma [5], which shows that the behavior of a controllable linear system can be fully characterized using data from a sufficiently excited trajectory. This result has led to a behavioral framework for data-driven control of linear systems [6], [7], with applications to linear quadratic regulation [6], [8] and model predictive control [9], and extensions to nonlinear systems [10] and switched systems [11].

Modern control systems are increasingly implemented over digital communication networks, where data must be transmitted through rate-limited channels [12]. This has motivated extensive research on control under data-rate constraints. A common approach is to quantize measurements into finite sets before transmission, thereby enabling finite data rates [13]. Foundational results using this approach are surveyed in [12], [14], with applications in areas such as power electronics, robotics, and aerospace [15]. Recent developments span robust control [16]–[18], event-triggered control [19], and switched systems [20], [21].

These considerations motivate the study of data-driven control under data-rate constraints. Most existing data-driven control methods rely on unquantized measurements, which limits their applicability in communication-constrained environments. Extensions to noisy data have introduced robust-

ness guarantees using tools such as semidefinite programming [8], the matrix S-lemma [22], and data informativity [23]. However, these approaches do not explicitly address data-rate constraints.

Data-driven control under quantized measurements remains relatively underexplored. In [24], stabilization of unknown linear systems with quantized control inputs is studied using noisy unquantized trajectory data. Related works consider superstabilization using partially quantized data [25], event-triggered control with quantized state transmission [26], and the trade-off between quantization densities for controller design and feedback stabilization [27]. These works employ static quantizers with infinitely many levels in the feedback loop, and therefore do not achieve finite data rates. Moreover, [24]–[26] rely at least partly on unquantized data to compute feedback gains, leaving a central challenge in data-driven control under data-rate constraints unresolved. The present paper addresses both issues.

We develop a data-driven framework for stabilizing unknown discrete-time linear systems using only quantized state measurements that take values in finite sets.

First, in the controller design stage, quantized data are collected under a prescribed quantization error bound and used to formulate a semidefinite program (SDP). We establish a verifiable condition under which any feasible solution to the SDP yields a stabilizing feedback gain, together with a Lyapunov matrix used for quantizer design in the stabilization stage. We further show that, under controllability and persistent excitation, the SDP is feasible and every optimal solution yields a stabilizing feedback gain when the quantization error is sufficiently small.

Second, in the stabilization stage, a Lyapunov-based quantizer update rule using a fixed number of quantization cells is developed. We establish that, when this number is sufficiently large, the closed-loop system under quantized state feedback converges exponentially to the origin.

In both stages, the number of quantization cells is finite, and it remains constant during stabilization. Thus, the proposed approach provides data-driven stabilization using only quantized state measurements under finite data rates.

The remainder of the paper is organized as follows. Section II introduces the system model, control objective, and quantization scheme. Section III develops data-driven controller design and establishes conditions for obtaining stabilizing feedback gains from quantized data. Section IV develops quantized feedback stabilization and establishes exponential convergence. Section V presents simulation results, and Section VI concludes the paper.

Notation: For a vector v , denote by $\|v\|$ and $\|v\|_\infty$ its

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Euclidean and infinity norms, respectively. For a matrix $M \in \mathbb{R}^{n \times m}$, denote by $\underline{\sigma}(M)$, $\|M\|$, and $\|M\|_F$ its smallest singular value, spectral norm (i.e., largest singular value), and Frobenius norm, respectively. For a symmetric matrix $S \in \mathbb{R}^{n \times n}$, denote by $\bar{\lambda}(S)$ and $\underline{\lambda}(S)$ its largest and smallest eigenvalues, respectively. For symmetric matrices $P, R \in \mathbb{R}^{n \times n}$, write $P \succ R$ and $P \succeq R$ when $P - R$ is positive definite and positive semidefinite, respectively.

II. PROBLEM FORMULATION

Consider a discrete-time linear time-invariant (LTI) system

$$x(k+1) = Ax(k) + Bu(k), \quad x(0) = x_0, \quad k \in \mathbb{Z}_{\geq 0}, \quad (1)$$

where $x(k) \in \mathbb{R}^n$ is the state and $u(k) \in \mathbb{R}^m$ is the control input. We assume that the dimensions n and m are known, but the matrices $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times m}$ are unknown. Instead, only upper bounds on their norms are known:

$$\|A\| \leq \alpha, \quad \|B\| \leq \beta. \quad (2)$$

Our goal is to stabilize the unknown LTI system (1) using quantized state measurements $\hat{x}(k) = q_k(x(k))$, where $q_k(\cdot)$ is a time-varying quantizer that maps $\mathbb{R}^n$ to a *finite* set of points at each time k .

The proposed approach consists of two stages. First, in the *controller design stage*, we collect input and quantized state data and use them to compute a stabilizing feedback gain. The quantizer update rule in this stage is designed to maintain a prescribed quantization error bound. Second, in the *stabilization stage*, we employ a Lyapunov-based quantizer update rule with a fixed number of quantization cells and establish exponential convergence under quantized state feedback. The controller design stage need not be performed offline. For concreteness, we consider stabilization beginning directly after controller design without resetting the system; the designed controller can also be applied after a system reset or at a later time.

The quantization scheme is introduced in Section II-A. The design and stabilization stages are developed in Sections III and IV, respectively.

A. Quantization scheme

The initial state x_0 is unknown. We assume that there exists a known bound $E_0 > 0$ such that

$$\|x_0\|_\infty \leq E_0.$$

If unavailable, such a bound can be obtained in finite time using search techniques in quantized control; see, e.g., [13].

At each time k , we maintain a bound E_k such that

$$\|x(k)\|_\infty \leq E_k, \quad (3)$$

that is, $x(k)$ lies in the hypercube

$$\mathcal{R}(E_k) := \{v \in \mathbb{R}^n : \|v\|_\infty \leq E_k\},$$

which serves as the quantization range. Given E_k , quantizer update rules are designed to determine E_{k+1} such that $x(k+1) \in \mathcal{R}(E_{k+1})$. The update rules differ between the two stages and are derived in Sections III-B and IV-A.

The quantizer partitions $\mathcal{R}(E_k)$ into N_k^n equal hypercubic cells, with N_k cells per dimension, where N_k is a positive integer. The quantized measurement $\hat{x}(k)$ is defined as the center of the cell containing $x(k)$. Consequently, the quantization error

$$d(k) := \hat{x}(k) - x(k)$$

satisfies

$$\|d(k)\|_\infty = \|\hat{x}(k) - x(k)\|_\infty \leq \frac{E_k}{N_k}. \quad (4)$$

Since the quantizer takes values in a set of cardinality N_k^n , the measurement at time k can be represented using a finite number of bits.

III. DATA-DRIVEN CONTROLLER DESIGN

A. Data-driven control preliminaries

During the controller design stage, we excite the unknown LTI system (1) with an input sequence $\{u(k)\}_{k=0}^{T-1}$ satisfying

$$\|u(k)\| \leq \bar{u}, \quad \forall k \in \{0, 1, \dots, T-1\}, \quad (5)$$

and collect the quantized state measurement sequence $\{\hat{x}(k)\}_{k=0}^T$. Define the corresponding data matrices

$$\begin{aligned} \hat{X}_0 &:= [\hat{x}(0) \quad \hat{x}(1) \quad \dots \quad \hat{x}(T-1)], \\ \hat{X}_1 &:= [\hat{x}(1) \quad \hat{x}(2) \quad \dots \quad \hat{x}(T)]. \end{aligned} \quad (6)$$

The data matrices $U_0 \in \mathbb{R}^{m \times T}$, $X_0, X_1 \in \mathbb{R}^{n \times T}$, and $D_0, D_1 \in \mathbb{R}^{n \times T}$ for the input $u(k)$, unquantized state $x(k)$, and quantization error $d(k)$, respectively, are defined similarly. The matrices X_0, X_1, D_0, D_1 are introduced only for analysis and are not available for control synthesis.

A key condition for computing a stabilizing controller is

$$\text{rank} \begin{bmatrix} U_0 \\ \hat{X}_0 \end{bmatrix} = n + m. \quad (7)$$

Following [5], an input sequence is *persistently exciting (PE)* of order L if its Hankel matrix of depth L has full row rank, and for a controllable LTI system, PE of order $n+1$ implies

$$\text{rank} \begin{bmatrix} U_0 \\ X_0 \end{bmatrix} = n + m. \quad (8)$$

We assume

$$T \geq m(n+1) + n, \quad (9)$$

which is necessary for $\{u(k)\}_{k=0}^{T-1}$ to be PE of order $n+1$. Under controllability and PE of order $n+1$, Theorem 2 below shows that (7) holds for sufficiently small quantization errors.

B. Quantized data collection

For $k < T$, using (2), (3), and (5), we obtain

$$\|x(k+1)\|_\infty \leq \|Ax(k) + Bu(k)\| \leq \alpha\sqrt{n}E_k + \beta\bar{u}.$$

To ensure that (3) holds for all $k \leq T$, we let

$$E_{k+1} := \alpha\sqrt{n}E_k + \beta\bar{u}, \quad k = 0, 1, \dots, T-1. \quad (10)$$

To maintain a prescribed quantization error bound $\delta > 0$, we let

$$N_k := \left\lceil \frac{E_k}{\delta} \right\rceil, \quad k = 0, 1, \dots, T, \quad (11)$$

where $\lceil \cdot \rceil$ is the ceiling function. Then (4) gives

$$\|d(k)\|_\infty \leq \delta \quad \forall k \in \{0, 1, \dots, T\}. \quad (12)$$

The update rule (10) yields a bound on $\|x(k+1)\|_\infty$ that is independent of the collected data but can lead to exponential growth of E_k and N_k during this stage. A tighter update rule can be obtained by exploiting the available quantized measurement $\hat{x}(k)$ and letting

$$E_{k+1} := \alpha \left(\|\hat{x}(k)\| + \frac{\sqrt{n}E_k}{N_k} \right) + \beta \bar{u}, \quad k = 0, 1, \dots, T-1. \quad (13)$$

Using (4), this update also ensures that (3) holds for all $k \leq T$ and can significantly reduce the required number of quantization cells.

Although the number of quantization cells may vary during this stage, E_k and N_k can be determined at each time from their update rules and, when (13) is used, previously transmitted quantized measurements; hence they need not be communicated separately. Thus, only the quantizer output at each time needs to be transmitted, and it can be represented using a finite number of bits. The number of quantization cells remains constant during the stabilization stage in Section IV.

C. Data-driven controller synthesis

Motivated by the optimization-based framework in [8], we formulate the following *semidefinite program* (SDP) using the input and quantized state data matrices:

$$\min_{(\gamma, Q, P, L, W)} \gamma \quad (14a)$$

subject to

$$\hat{X}_1 Q P^{-1} Q^\top \hat{X}_1^\top - P + I \preceq 0, \quad (14b)$$

$$P \succeq I, \quad (14c)$$

$$L - U_0 Q P^{-1} Q^\top U_0^\top \succeq 0, \quad (14d)$$

$$\hat{X}_0 Q = P, \quad (14e)$$

$$W - Q P^{-1} Q^\top \succeq 0, \quad (14f)$$

$$\text{tr}(P) + \text{tr}(L) + \omega \text{tr}(W) \leq \gamma. \quad (14g)$$

The additional variable W and parameter $\omega > 0$ introduce a regularization term motivated by robustness to disturbances; see [8, Sec. 5]. Larger values of ω encourage solutions with smaller $\text{tr}(Q P^{-1} Q^\top)$, which can make the sufficient condition in Theorem 1 below easier to satisfy.

The data matrices satisfy

$$X_1 = A X_0 + B U_0, \quad \hat{X}_0 = X_0 + D_0, \quad \hat{X}_1 = X_1 + D_1.$$

Hence

$$\hat{X}_1 = A X_0 + B U_0 + D_1 = A \hat{X}_0 + B U_0 + (D_1 - A D_0).$$

Let

$$\Delta_0 := D_1 - A D_0 \in \mathbb{R}^{n \times T},$$

which captures the discrepancy introduced by the quantized measurements. By (2) and (12), the columns of Δ_0 satisfy

$$\|d(k+1) - A d(k)\| \leq \sqrt{n}(\alpha+1)\delta \quad \forall k \in \{0, 1, \dots, T-1\}.$$

Hence

$$\|\Delta_0\| \leq \|\Delta_0\|_F \leq \sqrt{nT}(\alpha+1)\delta =: \hat{\delta}, \quad (15)$$

where $\hat{\delta}$ can be made arbitrarily small by decreasing δ , at the expense of requiring more quantization cells.

The following theorem establishes a verifiable condition under which any feasible solution to the SDP (14) yields a stabilizing controller.

Theorem 1. *Let $(\bar{\gamma}, \bar{Q}, \bar{P}, \bar{L}, \bar{W})$ be any feasible solution to (14), and define*

$$\bar{K} := U_0 \bar{Q} \bar{P}^{-1}, \quad \bar{M} := \bar{Q} \bar{P}^{-1} \bar{Q}^\top.$$

Then

$$(A + B \bar{K}) \bar{P} (A + B \bar{K})^\top - \bar{P} \preceq -\eta_1 I, \quad (16)$$

where

$$\eta_1 := 1 - \hat{\delta}^2 \|\bar{M}\| - 2\hat{\delta} \|\hat{X}_1 \bar{M}\|. \quad (17)$$

If

$$\eta_1 > 0, \quad (18)$$

then $A + B \bar{K}$ is Schur stable.

Proof. Since $\bar{P} = \hat{X}_0 \bar{Q}$ and $\bar{K} \bar{P} = U_0 \bar{Q}$,

$$(A + B \bar{K}) \bar{P} = (A \hat{X}_0 + B U_0) \bar{Q} = (\hat{X}_1 - \Delta_0) \bar{Q}.$$

Then

$$\begin{aligned} (A + B \bar{K}) \bar{P} (A + B \bar{K})^\top &= (\hat{X}_1 - \Delta_0) \bar{Q} \bar{P}^{-1} \bar{Q}^\top (\hat{X}_1 - \Delta_0)^\top = \hat{X}_1 \bar{M} \hat{X}_1^\top + \bar{\Psi}, \end{aligned}$$

where

$$\bar{\Psi} := \Delta_0 \bar{M} \Delta_0^\top - \hat{X}_1 \bar{M} \Delta_0^\top - \Delta_0 \bar{M} \hat{X}_1^\top.$$

Combining this with (14b) yields

$$(A + B \bar{K}) \bar{P} (A + B \bar{K})^\top - \bar{P} \preceq \bar{\Psi} - I.$$

Using (15), we obtain

$$\|\bar{\Psi}\| \leq \hat{\delta}^2 \|\bar{M}\| + 2\hat{\delta} \|\hat{X}_1 \bar{M}\| = 1 - \eta_1.$$

Since $\bar{\Psi}$ is symmetric,

$$\bar{\Psi} \preceq \|\bar{\Psi}\| I \preceq (1 - \eta_1) I,$$

and thus (16) holds. If $\eta_1 > 0$, then (16) implies

$$(A + B \bar{K}) \bar{P} (A + B \bar{K})^\top - \bar{P} \prec 0,$$

where $\bar{P} \succ 0$. Therefore, $A + B \bar{K}$ is Schur stable. $\square$

In practice, the feedback gain $\bar{K}$ can be computed by solving (14) using the available input and quantized state data. The sufficient condition (18) can also be verified directly from these data.

Our second theorem establishes conditions under which the SDP (14) is feasible and every optimal solution yields a stabilizing controller. Its proof, omitted due to space limitations, uses the robust fundamental lemma in [28] and can be found in [29].

Theorem 2. Suppose that the LTI system (1) is controllable and the input sequence $\{u(k)\}_{k=0}^{T-1}$ is PE of order $n+1$. There exists $\delta^* > 0$ such that, if the quantization error bound $\delta < \delta^*$, then the rank condition (7) holds, the SDP (14) is feasible, and every optimal solution to (14) yields a stabilizing controller.

Theorem 2 is restricted to optimal solutions to the SDP (14), in contrast to Theorem 1, which applies to all feasible solutions. This restriction arises because the proof relies on comparing the optimal value of (14) with the objective value of a specifically constructed feasible solution.

An explicit sufficient bound on δ is provided in the proof. It depends on the unknown matrices A and B and therefore cannot be verified from the available data. Nevertheless, the result provides an existence guarantee: a stabilizing controller can be obtained under sufficiently fine quantization.

IV. QUANTIZED FEEDBACK STABILIZATION

Having obtained a stabilizing feedback gain from quantized data, we now address closed-loop stabilization using quantized state feedback.

A. Quantized state feedback implementation

At time T , suppose that a feasible solution $(\bar{\gamma}, \bar{Q}, \bar{P}, \bar{L}, \bar{W})$ to the SDP (14) has been obtained and satisfies (18). Then by Theorem 1, the feedback gain $\bar{K} := U_0 \bar{Q} \bar{P}^{-1}$ satisfies (16), and $A + B\bar{K}$ is Schur stable.

For the transition between the two stages, we let $u(T) = 0$ and begin quantized state feedback at time $T+1$. During the stabilization stage, the quantizer uses a fixed number of quantization cells and exponentially decaying bounds:

$$N_k = N_s, \quad E_{k+1} = \mu E_k, \quad k = T+1, T+2, \dots, \quad (19)$$

where the initial bound E_{T+1} , the contraction factor μ , and a sufficient condition on N_s for $\mu \in (0, 1)$ are derived in Section IV-B. With these choices, the closed-loop state under

$$x(k+1) = Ax(k) + B\bar{K}\hat{x}(k), \quad k \geq T+1, \quad (20)$$

remains in the quantization range (i.e., (3) holds) for all $k \geq T+1$ and converges exponentially to the origin.

B. Exponential convergence

We establish exponential convergence by constructing nested sets that contract under the closed-loop dynamics, based on sublevel sets of

$$V(x) := x^\top \bar{P}^{-1} x, \quad (21)$$

where $\bar{P} \succ 0$ is from the feasible solution to (14) obtained at time T .

We first construct a sublevel set of V containing $x(T+1)$ and derive E_{T+1} such that the quantization range $\mathcal{R}(E_{T+1})$ contains this set. Recall that

$$\mathcal{R}(E) = \{x \in \mathbb{R}^n : \|x\|_\infty \leq E\},$$

and define the sublevel set

$$\mathcal{S}(E) := \left\{x \in \mathbb{R}^n : V(x) \leq \frac{E^2}{\bar{\lambda}(\bar{P})}\right\}.$$

Since

$$\frac{\|x\|_\infty^2}{\bar{\lambda}(\bar{P})} \leq \frac{\|x\|^2}{\bar{\lambda}(\bar{P})} \leq V(x) \leq \frac{\|x\|^2}{\underline{\lambda}(\bar{P})} \leq \frac{n\|x\|_\infty^2}{\underline{\lambda}(\bar{P})} \quad \forall x \in \mathbb{R}^n,$$

we have

$$\mathcal{S}(E) \subset \mathcal{R}(E) \subset \mathcal{S}\left(\sqrt{\frac{n\bar{\lambda}(\bar{P})}{\underline{\lambda}(\bar{P})}} E\right) \quad \forall E > 0.$$

Using $u(T) = 0$ and $x(T) \in \mathcal{R}(E_T)$, we obtain

$$\|x(T+1)\|_\infty \leq \|Ax(T)\| \leq \alpha\|x(T)\| \leq \alpha\sqrt{n}E_T.$$

Therefore, defining

$$E_{T+1} := \alpha n \sqrt{\frac{\bar{\lambda}(\bar{P})}{\underline{\lambda}(\bar{P})}} E_T \quad (22)$$

yields $x(T+1) \in \mathcal{R}(\alpha\sqrt{n}E_T) \subset \mathcal{S}(E_{T+1}) \subset \mathcal{R}(E_{T+1})$. Thus, $\mathcal{R}(E_{T+1})$ can be used as the initial quantization range for the stabilization stage.

A tighter initial bound can be obtained by exploiting the available quantized measurement $\hat{x}(T)$ and letting

$$E_{T+1} := \alpha \sqrt{\frac{n\bar{\lambda}(\bar{P})}{\underline{\lambda}(\bar{P})}} \left(\|\hat{x}(T)\| + \frac{\sqrt{n}E_T}{N_T} \right), \quad (23)$$

which also ensures $x(T+1) \in \mathcal{S}(E_{T+1}) \subset \mathcal{R}(E_{T+1})$.

Theorem 3. Let $\varepsilon > 0$. Suppose that the number of quantization cells satisfies

$$N_s > \sqrt{\frac{n(\theta + \sqrt{\phi + \varepsilon})^2 \bar{\lambda}(\bar{P})^3}{\eta_1 \underline{\lambda}(\bar{P})} - n\varepsilon \bar{\lambda}(\bar{P})} =: N_*, \quad (24)$$

where

$$\begin{aligned} \theta &:= \frac{\beta(\alpha + \beta\|\bar{K}\|)\|\bar{K}\|\bar{\lambda}(\bar{P})}{\sqrt{\eta_1} \underline{\lambda}(\bar{P})}, \\ \phi &:= \theta^2 + \frac{\beta^2\|\bar{K}\|^2}{\underline{\lambda}(\bar{P})}, \end{aligned} \quad (25)$$

$\eta_1 \in (0, 1)$ is defined in (17), and $\alpha, \beta > 0$ are as in (2). Then

$$\|x(k)\|_\infty \leq \mu^{k-T-1} E_{T+1} \quad \forall k \geq T+1, \quad (26)$$

where

$$\mu := \sqrt{\max\left\{\frac{N_*^2}{N_s^2}, 1 - \frac{n\varepsilon \bar{\lambda}(\bar{P})}{N_s^2}\right\}} < 1. \quad (27)$$

Note that N_* in (24) is well defined and positive for every $\varepsilon > 0$ since $\bar{P} \succeq I$, $\eta_1 \in (0, 1)$, and $\theta, \phi \geq 0$.

The proof of Theorem 3 relies on the following lemma.

Lemma 1. Suppose that at time $k \geq T+1$,

$$x(k) \in \mathcal{S}(E_k) \subset \mathcal{R}(E_k),$$

where $\mathcal{R}(E_k)$ is the quantization range. Then

$$x(k+1) \in \mathcal{S}(\mu E_k) \subset \mathcal{R}(\mu E_k), \quad (28)$$

where μ is defined in (27).

Proof. For brevity, let $x = x(k)$, $x^+ = x(k+1)$, $V = V(x)$, $V^+ = V(x^+)$, $d = d(k)$, and $E = E_k$ in this proof. The closed-loop dynamics (20) gives

$$\begin{aligned} V^+ &= ((A + B\bar{K})x + B\bar{K}d)^\top \bar{P}^{-1}((A + B\bar{K})x + B\bar{K}d) \\ &= x^\top (A + B\bar{K})^\top \bar{P}^{-1}(A + B\bar{K})x \\ &\quad + 2x^\top (A + B\bar{K})^\top \bar{P}^{-1}B\bar{K}d + d^\top \bar{K}^\top B^\top \bar{P}^{-1}B\bar{K}d. \end{aligned}$$

We first bound the nominal closed-loop term. By (16),

$$(A + B\bar{K})\bar{P}(A + B\bar{K})^\top \preceq \left(1 - \frac{\eta_1}{\bar{\lambda}(\bar{P})}\right)\bar{P}.$$

One can then show that

$$(A + B\bar{K})^\top \bar{P}^{-1}(A + B\bar{K}) \preceq \left(1 - \frac{\eta_1}{\bar{\lambda}(\bar{P})}\right)\bar{P}^{-1}.$$

Consequently,

$$x^\top (A + B\bar{K})^\top \bar{P}^{-1}(A + B\bar{K})x \leq V - \frac{\eta_1}{\bar{\lambda}(\bar{P})^2} \|x\|^2.$$

We next bound the terms due to quantization. Using (2) and (4), we obtain

$$\begin{aligned} x^\top (A + B\bar{K})^\top \bar{P}^{-1}B\bar{K}d &\leq \frac{\beta(\alpha + \beta\|\bar{K}\|)\|\bar{K}\|}{\bar{\lambda}(\bar{P})} \frac{\sqrt{n}E}{N_s} \|x\| \\ &= \frac{\theta\sqrt{n}E}{N_s} \frac{\sqrt{\eta_1}}{\bar{\lambda}(\bar{P})} \|x\|, \end{aligned}$$

and

$$d^\top \bar{K}^\top B^\top \bar{P}^{-1}B\bar{K}d \leq \frac{\beta^2\|\bar{K}\|^2}{\bar{\lambda}(\bar{P})} n\|d\|_\infty^2 \leq \frac{\beta^2\|\bar{K}\|^2}{\bar{\lambda}(\bar{P})} \frac{nE^2}{N_s^2}.$$

Combining these bounds and completing the square yields

$$V^+ \leq V - \left(\frac{\sqrt{\eta_1}}{\bar{\lambda}(\bar{P})} \|x\| - \frac{\theta\sqrt{n}E}{N_s}\right)^2 + \frac{\phi nE^2}{N_s^2}. \quad (29)$$

We consider two cases according to the magnitude of the state. If

$$\|x\| \geq \frac{\bar{\lambda}(\bar{P})}{\sqrt{\eta_1}} \frac{(\theta + \sqrt{\phi + \varepsilon})\sqrt{n}E}{N_s}, \quad (30)$$

then the squared term in (29) is at least $(\phi + \varepsilon)nE^2/N_s^2$. Since $x \in \mathcal{S}(E)$, we have

$$V^+ \leq V - \frac{\varepsilon nE^2}{N_s^2} \leq \left(1 - \frac{n\varepsilon\bar{\lambda}(\bar{P})}{N_s^2}\right) \frac{E^2}{\bar{\lambda}(\bar{P})}.$$

Otherwise, using $V \leq \|x\|^2/\bar{\lambda}(\bar{P})$ in (29) gives

$$\begin{aligned} V^+ &\leq \left(\frac{1}{\bar{\lambda}(\bar{P})} - \frac{\eta_1}{\bar{\lambda}(\bar{P})^2}\right) \|x\|^2 + \frac{2\theta\sqrt{n}E}{N_s} \frac{\sqrt{\eta_1}}{\bar{\lambda}(\bar{P})} \|x\| \\ &\quad + \frac{\beta^2\|\bar{K}\|^2}{\bar{\lambda}(\bar{P})} \frac{nE^2}{N_s^2}. \end{aligned}$$

Since $\bar{\lambda}(\bar{P}) \geq \underline{\lambda}(\bar{P}) \geq 1 > \eta_1$, the right-hand side is nondecreasing in $\|x\|$. Using the upper bound on $\|x\|$ given by the complement of (30), we obtain

$$V^+ < \frac{\bar{\lambda}(\bar{P})^2}{\eta_1 \bar{\lambda}(\bar{P})} \frac{(\theta + \sqrt{\phi + \varepsilon})^2 nE^2}{N_s^2} - \frac{\varepsilon nE^2}{N_s^2} = \frac{N_s^2}{N_s^2} \frac{E^2}{\bar{\lambda}(\bar{P})}.$$

Combining both cases, we obtain

$$V^+ \leq \frac{\mu^2 E^2}{\bar{\lambda}(\bar{P})}.$$

Hence $x^+ \in \mathcal{S}(\mu E) \subset \mathcal{R}(\mu E)$. $\square$

Proof of Theorem 3. By either (22) or (23), $x(T+1) \in \mathcal{S}(E_{T+1}) \subset \mathcal{R}(E_{T+1})$. Therefore, iterating Lemma 1 gives

$$x(k) \in \mathcal{R}(\mu^{k-T-1} E_{T+1}) \quad \forall k \geq T+1,$$

which implies (26). Under (24), both terms in the maximum in (27) are strictly smaller than 1 and $N_* > 0$. Hence $\mu \in (0, 1)$. $\square$

V. SIMULATION RESULTS

We illustrate the proposed approach on the LTI system (1) with

$$A = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

which is unstable in open loop. The dimensions are $n = 2$ and $m = 1$. We use $T = m(n+1)+n = 5$, the minimum data length allowed by (9), for the design stage. The stabilization stage is simulated for 30 steps.

During the design stage, the input sequence is PE of order $n+1 = 3$ and satisfies $\|u(k)\| \leq \bar{u} = 1$. The initial state is $x_0 = [0.4 \ 0.4]^\top$. We use $E_0 = 1$, $\delta = 0.01$, $\alpha = 3 > \|A\|$, and $\beta = 1.5 > \|B\|$. Using (11) together with the measurement-dependent update rule (13), the number of quantization cells per dimension is

$$(N_0, N_1, \dots, N_5) = (100, 198, 525, 213, 382, 561).$$

The collected data satisfy the rank condition (7). Solving the quantized SDP (14) with $\omega = 1$ yields

$$\bar{K} = U_0 \bar{Q} \bar{P}^{-1} = [-1.089 \quad -0.277],$$

and $\eta_1 = 0.122 > 0$. Thus, the verifiable condition (18) holds, and Theorem 1 guarantees that $A + B\bar{K}$ is Schur stable. Indeed, its eigenvalues are 0.392 and -0.481 .

During the stabilization stage, the initial quantization range is computed using the measurement-dependent initialization (23), yielding $E_{T+1} = 19.569$. The sufficient condition (24) in Theorem 3 gives the conservative bound $N_s > N_* = 8524$. In the simulation, we use only $N_s = 50$ cells per dimension and set $\mu = 0.95$. Although these choices are not certified by Theorem 3, the simulation results below show successful stabilization. One source of conservativeness is the mismatch between the hypercubic quantization ranges and the Lyapunov sublevel sets used in the analysis. Quantization regions better aligned with $\bar{P}$ may therefore reduce the required number of cells.

Fig. 1(a) shows the state norm $\|x(k)\|_\infty$ and the quantization bound E_k . During the design stage ($k = 0, \dots, 5$), the PE input excites the system while the quantizer update rule adapts the quantization range to maintain the prescribed quantization error bound. At the transition to stabilization, the bound increases to $E_{T+1} = 19.569$ according to (23).

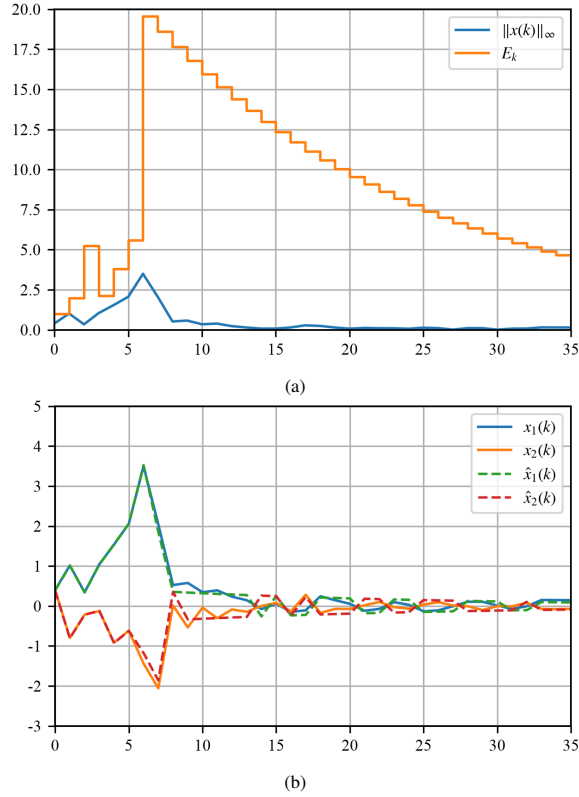


Fig. 1: Evolution of (a) the state norm $\|x(k)\|_\infty$ and the quantization bound E_k , and (b) the state components $x_i(k)$ and their quantized measurements $\hat{x}_i(k)$. The design stage ends at $k = 5$, and quantized state feedback begins at $k = 6$.

Thereafter, E_k decreases exponentially with contraction factor $\mu = 0.95$, while the state remains within the quantization range and converges to the origin.

Fig. 1(b) shows the state components $x_i(k)$ and their quantized measurements $\hat{x}_i(k)$. The quantized measurements closely track the states during data collection, and both converge during stabilization.

VI. CONCLUSION

This paper developed a data-driven framework for stabilizing unknown discrete-time linear systems using only quantized state measurements that take values in finite sets. The proposed approach enables both controller design and closed-loop stabilization under finite data rates.

Future research directions include reducing the conservativeness of the quantizer design and extending the proposed framework to continuous-time LTI systems and switched systems.

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